

The Big Upside

You just spent a whole section on what goes wrong. Hallucinations, bias, flattery, fakes. All of it real, all of it worth knowing. Now here is the other half, which is just as real: this same technology solved a problem that had beaten biology for fifty years.

A 50-YEAR-OLD PROBLEM

Your body runs on proteins. Think of them as tiny machines. They digest your food, carry oxygen, fight off infections, and build muscle. Each one starts as a long string of chemical beads, and then that string folds itself into a specific 3D shape in a fraction of a second.

The shape is the whole thing. What a protein does depends entirely on how it folds. Which means how a disease works comes down to shapes, and whether a drug can block it comes down to shapes.

Reading the string was easy. Predicting the shape it folds into was brutal. One protein could take a scientist years of lab work and a pile of expensive equipment. In fifty years of trying, humans had worked out roughly 200,000 shapes. There are hundreds of millions of proteins.

 **Why it's so hard:** A medium-sized protein has more possible shapes than there are atoms in the universe. It finds the right one almost instantly, almost every time. Scientists named this Levinthal's paradox and stared at it for decades.

THE GUY WHO SOLVED IT

Demis Hassabis hit chess master level at 13. At 17 he was co-designing a video game called Theme Park that sold millions of copies. Then he went and got a PhD in neuroscience, because he wanted to understand how the brain works well enough to build one. In 2010 he started an AI company called DeepMind. Google bought it four years later.

In 2020 his team entered a system called AlphaFold into CASP, the contest scientists run every two years to see who can predict protein shapes best. AlphaFold's predictions came back about as accurate as real lab experiments. The organizers said the fifty-year problem was essentially solved.

Then DeepMind did the part that actually mattered. In 2022 they released the predicted shape of around 200 million proteins, close to every protein known to science, free to anybody who wants them. More than two million users in 190 countries have pulled from that database. In 2024, Hassabis and his colleague John Jumper shared the Nobel Prize in Chemistry for it.

Fifty years, then two.



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I've dedicated my career to advancing AI because of its unparalleled potential to improve the lives of billions of people.

— DEMIS HASSABIS · On winning the 2024 Nobel Prize in Chemistry

IT'S NOT JUST PROTEINS

Same pattern each time. A problem with too many possibilities for humans to search by hand, and a model that can search it anyway.

MEDICINE

New antibiotics

Bacteria are learning to survive our drugs faster than we find new ones. In 2023, MIT researchers used AI to screen thousands of compounds and found one that kills a superbug that shrugs off almost everything else. They named it abaucin.

WEATHER

Weather you can plan around

DeepMind built a weather model that runs a 10-day global forecast in about a minute on a single machine. The old way takes hours on a supercomputer. Better storm warnings means more time to get people out.

MATERIALS

Materials that don't exist yet

Better batteries, solar panels, and chips all wait on someone finding the right crystal. DeepMind's GNoME predicted about 2.2 million new ones, including roughly 380,000 stable enough to be worth making. Researchers called it centuries of discovery, done at once.

Notice that none of this is a chatbot. Same core idea you learned in Understand AI, which is a model finding patterns in enormous amounts of data, just pointed at a different kind of problem. “AI” is a much bigger thing than the app on your phone.

NOW THE HONEST PART

Hassabis thinks this is only the start. He founded a second company, Isomorphic Labs, to use this kind of AI to design actual medicines, and he has said he believes AI can help cure all disease within about a decade.

That is a prediction, not a result. Knowing a protein's shape is not the same as having a drug. A medicine still has to be designed, tested in a dish, tested in animals, tested in people across three rounds of trials, and approved by regulators. Isomorphic's first candidates are only now reaching human trials, and that timeline has already slipped once. Ten years is aggressive.

See what just happened there. You gave the good news exactly the same treatment you gave the bad news in the last section. That is the skill. Hype comes in both flavors, and “AI will fix everything” deserves the same check as “AI will ruin everything.”

 **The upside is real, and it's checkable.** AlphaFold already happened. Curing all disease is still a prediction. Learn to tell those two apart and you can be genuinely excited about AI without being sold anything.

The upside is real, and it already happened.

Judge the good news like you judge the bad.